For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PURE MATHEMATICS **6042/1**

PAPER 1 3 Hours

SPECIMEN PAPER

Additional materials:
Answer booklet
Graph paper
List of Formulae MF7
Scientific calculator [Non-Programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 120.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

Answer **all** questions.

This specimen paper consists of 7 printed pages.

[Turn over

Answer **all** questions.

1. It is given that

$$f(x) = -3x^2 + 6x - 5.$$

(a) Express $f(x)$ in the form
$a(x+b)^2 + c$,

where a , b and c are constants to be found. [3]

(b) Hence state the coordinates of the turning point of $f(x)$. [1]

2. A geometric series has first term 5 and common ratio 0.75.
The sum of the first n terms of the series is denoted by S_n and the sum to infinity is denoted by S_∞.

Calculate the least value of n for which $S_\infty - S_n < 2$. [5]

3. Evaluate $\int_0^2 x^2 e^x \, dx$. [5]

4. The line $x + 3y = 1$, meets the curve

$$x^2 - xy + y^2 = 21$$

at points A and B.

Find the coordinates of A and B. [6]

5. (a) Prove the identity

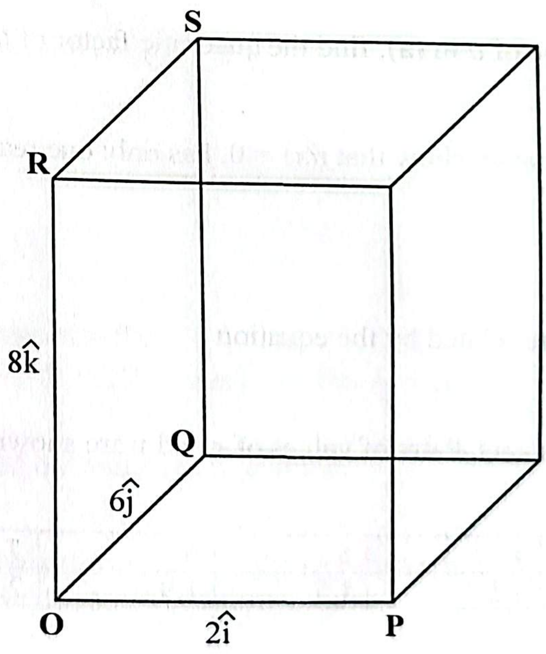
$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} \equiv 2\,\mathrm{cosec}\,\theta.$$ [3]

(b) Hence or otherwise solve the equation

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 4.5,$$

for $0° \leqslant \theta \leqslant 360°$. [4]

6.



The diagram shows a rectangular box such that

$\overrightarrow{OP} = 2\hat{\imath}$, $\overrightarrow{OQ} = 6\hat{\jmath}$ and $\overrightarrow{OR} = 8\hat{k}$

(a) **(i)** Find the position vector of the midpoint M of QS. [2]

(ii) Find a unit vector in the direction of the vector $\overrightarrow{PM}$ [3]

(b) The point X inside the box has position vector

$$2\hat{i} + 4\hat{j} + 6\hat{k}\cdot$$

Find the angle XOP to the nearest degree. [3]

7.

$$f(x) = 2x^3 - ax^2 + 3x - 1$$

is exactly divisible by $x - 1$.

(a) Find the value of *a*. [2]

(b) Using the value of *a* in **(a)**, find the quadratic factor of *f(x)*. [2]

(c) Hence or otherwise, show that *f(x)* = 0, has only one real root. [3]

8. The variables *x* and *y* are related by the equation
$y = a\sqrt{x} - b$,

where *a* and *b*, are constants. Pairs of values of *x* and *y* are shown in the table.

x	3	4	5	6	7
y	1.2	1.5	1.7	1.9	2.1

(a) Plot the graph of *y* against $\sqrt{x}$. [4]

(b) Use the graph to find the value of *a* and the value of *b*. [3]

9. Two complex numbers are denoted by
$a = 3 - 4i$ and $b = \sqrt{3} + 2i$.

(a) State the conjugate of *a*. [1]

(b) Find in the form, $x + iy$,

(i) ab, [2]

(ii) $\frac{a}{b}$. [2]

(c) Find the modulus and argument of *ab*. [4]

10. **(a)** On the same axes, sketch the graphs of

$y = |x^2 - 4x|$ and $y = x$. [3]

(b) Hence solve $|x^2 - 4x| > x$. [4]

11. **(a)** Find the first three terms in the expansion of

$(2 + ax)^5$. [3]

(b) **(i)** Given also that
$(b + 2x)(2 + ax)^5 \equiv 96 - 176x + cx^2$,

find the values of *a*, *b* and c. [6]

(ii) State the validity of the expansion in **b(i)** with the values obtained. [1]

12. A certain substance is formed in a chemical reaction. The mass of substance formed t seconds after the start of the reaction is x grams.

At any time t, the rate of formation of the substance is proportional to

$\left(100 - \frac{1}{2}x\right)$ when $t = 0, x = 0,$ and $\frac{dx}{dt} = 1.$

(a) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0{,}01\left(100 - \frac{1}{2}x\right)$$ [2]

(b) Solve the differential equation in **(i)**. [7]

(c) Find x when $t = 5$, giving the answer to 1 decimal place. [2]

(d) State what happens to the value of x as t becomes very large. [1]

13. **(a)** Express

$$\frac{2x^2}{(x-3)(x^2-2x-3)}$$ in partial fractions. [5]

(b) Hence or otherwise evaluate,

$$\int_4^5 \frac{2x^2}{(x-3)(x^2-2x-3)}dx$$ [5]

14. **(a)** State any **two** conditions for the Maclaurin's series to be valid for a given function. [2]

(b) **(i)** If $y = e^x \sin x$ shows that

$$\frac{d^2y}{dx^2} = 2\left(\frac{dy}{dx} - y\right).$$ [4]

(ii) By further differentiation of the result in part **(i)**, find the Maclaurin's expansion of the function $y = e^x \sin x$ as far as the term in x^3 [6]

15. (a) Show by graphical argument that the equation

$$\sin x + x - 1 = 0$$

has only one root for $0 \leq x \leq \frac{\pi}{2}$. [4]

(b) Verify that the root of the equation $\sin x + x - 1 = 0$ lies between 0 and $\frac{\pi}{2}$. [3]

(c) Use the Newton-Raphson method twice to find the root correct to 3 decimal places starting with $x_0 = 0{,}5$. [4]